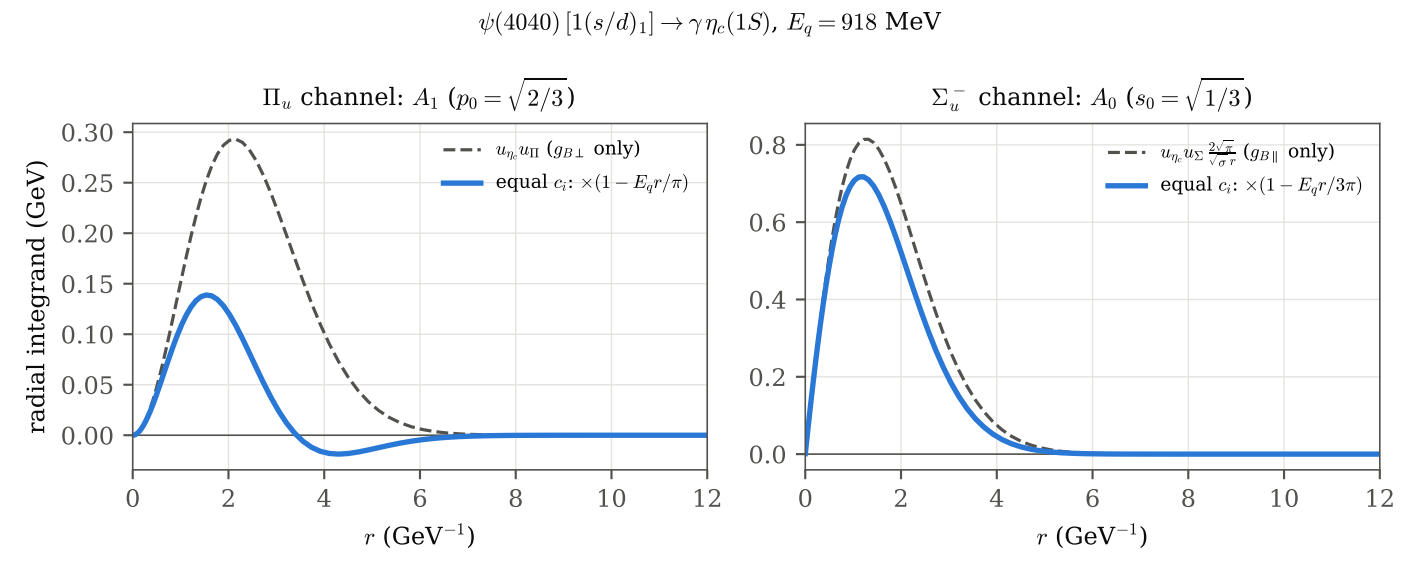
$\psi(4040)\,[1(s/d)_1] \to \gamma\,\eta_c(1S)$, $E_q = 918$ MeV
$\Pi_u$ channel: $A_1$ ($p_0 = \sqrt{2/3}$)
$u_{\eta_c} u_\Pi$ ($g_{B\perp}$ only)
equal $c_i$: $\times(1 - E_q r/\pi)$
radial integrand (GeV)
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0
2
4
6
8
10
12
$r$ (GeV$^{-1}$)
$\Sigma_u^-$ channel: $A_0$ ($s_0 = \sqrt{1/3}$)
$u_{\eta_c} u_\Sigma \frac{2\sqrt{\pi}}{\sqrt{\sigma}\, r}$ ($g_{B\parallel}$ only)
equal $c_i$: $\times(1 - E_q r/3\pi)$
0.8
0.6
0.4
0.2
0.0
0
2
4
6
8
10
12
$r$ (GeV$^{-1}$)